

Theory of Spectral Relativity — Version 3

Annex: Experimental Validation and Rigidity Observation

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1. Experimental Validation on DNS Data

The RST predicts that the sigmoidal shape parameter α of the normalised spectral energy flux $\Pi(k)/\epsilon$ is independent of Reynolds number. This section documents the experimental validation of this prediction on two independent DNS datasets from the Johns Hopkins Turbulence Database (JHTDB).

1.1 Protocol

For each dataset, the velocity field was sampled on a regular grid covering the full domain $[0, 2\pi]^3$. The 3D energy spectrum $E(k)$ was computed by spherical binning of the FFT, the cumulative flux $\Pi(k)$ was normalised by its maximum, and a sigmoidal fit $\Pi_{\text{norm}}(k) = (1 + \tanh(\alpha \cdot (\log k - \log k_c))) / 2$ was applied. The parameter α is the RST shape parameter. The same protocol was applied identically to both datasets.

1.2 Results

Dataset	Re_lambda	alpha	Uncertainty	RMS	Source
isotropic1024coarse	433	0.9308	+/- 0.0002	0.016	SOAP API, 4 snapshots
isotropic8192	~1256.8	0.9472	+/- 0.0419	0.0039	SciServer Compute (ceph), 16^3 chunks

The relative deviation between the two measurements is 1.76%, well within the predicted 5% tolerance. This confirms Reynolds-number independence of α across a factor of 3 in Re_λ (433 to 1256).

1.3 Physical parameters of isotropic8192

$$\nu = 4.385\text{e-}5, \text{Re}_\lambda = 1256.8, \epsilon = 1.3668$$

Data accessed via SciServer Compute with the Turbulence (ceph) volume mounted. Grid: $16^3 = 4096$ chunks of 64^3 points, evenly spaced across the full domain. Token: com.protonmail.pg68110-af32ac6e (provided by Ariel Lubonja, JHU).

2. Rigidity Observation of the Central Object

Beyond the experimental validation, the RST research programme has produced a structural observation: the central object $\Phi(x) = \tanh(\alpha \cdot x)$ has been independently recovered across 14 distinct mathematical and physical frameworks. None of these frameworks assumed it; each

reconstructed it.

2.1 The Central Object

$\Phi(x) = \tanh(\alpha \cdot x)$ [kink profile]

$\rho(x) = \alpha \cdot \operatorname{sech}^2(\alpha \cdot x)$ [spectral density]

$\psi_0(x) = \operatorname{sech}^2(\alpha \cdot x)$ [Goldstone mode]

2.2 The 14 Frameworks of Reconstruction

1. KPP / Reaction-diffusion

Test: Fixed point of the coupled KPP flow

Result: $\rho^* = \operatorname{sech}^2(\alpha \cdot x)$ exactly

2. Calculus of variations

Test: Unique minimiser of $A[\Phi]$ with symmetric double-well

Result: $\Phi = \tanh$, unique monotone finite-action solution

3. Integrable ODE / Bogomolny

Test: Autonomous first-order equation $\Phi' = f(\Phi)$

Result: $f(\Phi) = \alpha(1-\Phi^2)$, $\tanh$ by quadrature

4. Spectral theory / Poschl-Teller

Test: Hessian of $A[\Phi]$ around the kink

Result: $L = -d^2/dx^2 + 4\alpha^2 - 6\alpha^2 \operatorname{sech}^2(\alpha \cdot x)$

5. Sturm-Liouville theory

Test: Positive nodeless zero mode of L

Result: $\psi_0 = \operatorname{sech}^2$: unique nodeless ground state

6. Darboux factorisation / SUSY-QM

Test: Reconstruction $L \rightarrow \Phi \rightarrow V$ from the zero mode

Result: $\psi_0 = \operatorname{sech}^2 \Rightarrow \Phi = \tanh \Rightarrow$ quartic V

7. Morse theory

Test: Morse index of L

Result: Index = 0: stable kink, spectral gap = $4\alpha^2$

8. Differential geometry

Test: Heteroclinic connection in the phase plane (Φ, ρ)

Result: Unique orbit from $(-1,0)$ to $(+1,0)$ on $H=0$

9. Pohozaev identities

Test: Universal integrals over the heteroclinic trajectory

Result: $I_1/I_2 = 5/4$ (pure invariant $J_2 = 6/5$)

10. Dynamical systems

Test: Gradient flow fixed point attractor

Result: Unique monotone heteroclinic connection

11. Riemannian spectral geometry

Test: Metric $ds^2 = d\omega^2/\rho$ on spectral manifold

Result: Flat geodesic, isometric to H^2

12. Yang-Mills / QCD

Test: IR propagator, string tension, critical temperature

Result: $\sigma = 0.194 \text{ GeV}^2$ ($<0.5\%$ lattice), $T_c = 156.7 \text{ MeV}$ (1.1%)

13. Navier-Stokes / Turbulence

Test: Spectral flux $\Pi(k)$, JHTDB at $Re_\lambda = 433$ and 1256

Result: $\alpha = 0.93-0.95$, $RMS < 2\%$, Reynolds-independent

14. Pure invariants

Test: Constants from the kink profile alone

Result: $J_1=6$, $J_2=6/5$, $J_3=1/\sqrt{3}$ — universal, no free parameters

3. Status Note

The experimental validation (Section 1) is reproducible from the JHTDB public datasets using the SciServer Compute platform. The mathematical results (Section 2, frameworks 1-11 and 14) are verifiable via the V3 scripts or by direct computation. The physical validations (frameworks 12-13) are reproducible from public lattice QCD data and JHTDB.

A formal characterisation theorem — demonstrating that these 14 representations are exhaustive and necessarily reconstruct the same object — remains an open problem. The accumulated evidence documented here constitutes its empirical motivation.